

Bruno Avritzer

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Active Top Secret Clearance

EDUCATION

University of Southern California (advisor: Prof. Todd Brun) **2024**

Ph.D., Physics

Thesis: Quantum and Classical Steganography in Optical Systems

Cornell University **2019**

B.S., Engineering Physics

PROFESSIONAL EXPERIENCE & SERVICE

Leidos, Arlington VA **2024 - present**

Quantum Theory Group Lead, Staff Quantum Information Theorist

- Oversaw \$1.2m quantum communications/processing area IRAD portfolio, developing and patenting differentiating technologies in e.g. distributed quantum compilation, leading research team of 5, guiding and performing technical work, and presenting documentation for a nontechnical audience
- Developed, simulated, and presented to DARPA on novel beyond-QKD quantum communication methods and network applications, provided an interface between hardware, software, and theory areas for collaborative development
- Shaped and supported engagements and led capture efforts, building research teams, supporting \$1.1M in wins and serving in capture PI (DARPA HARQ) and technical lead roles
- Developed projects spanning areas from agentic AI to behavioral psychology

Quantum Economic Development Consortium (QED-C), Arlington VA **2026 - present**

Vice-Chairman, Standards and Performance Metrics TAC

- Organized webinars on quantum advantage and formed standardization working groups
- Represented QED-C in external speaking engagements (e.g. with Bitkom, INTC)

SELECTED PUBLICATIONS

Clayton, C., & **Avritzer, B.** (2025). Distributed Quantum Error Correction with Permutation-Invariant Approximate Codes. *arXiv 2509.25093*. Presented at QEC25, in review at *NPJ Quantum Information*.

Avritzer, B., & Brun, T. (2024). Quantum steganography via coherent- and Fock-state encoding in an optical medium. *Phys. Rev. A*, 109, 032401.

Weinstein, H., **Avritzer, B.**, Kinzogl, C., Brun, T., and Habif, J. (2025). High Fidelity Artificial Quantum Thermal State Generation using Encoded Coherent States. *IEEE Transactions on Quantum Engineering*, vol. 6, pp. 1-7, 2025, Art no. 5100307.

SKILLS

Programming languages: Python, Matlab, Java, Javascript, C#, C++, Bash, OCaml

Computer software/ frameworks: Scikit-learn, Pytorch, Numpy, StrawberryFields, Qiskit, Tket, CudaQ, MQT, Stim, BQSKit, Mathematica, Fusion360, AutoCAD, Unity, Arduino, Slurm (HPC)

Publications

Avritzer, B., & Sankary, N. (2026). Near-Term Reduction in Nonlocal Gate Count from Distributed Logical Qubits. *arXiv 2604.21722*.

Clayton, C., & **Avritzer, B.** (2025). Distributed Quantum Error Correction with Permutation-Invariant Approximate Codes. *arXiv 2509.25093*. *In review at NPJ Quantum Information*.

Avritzer, B., & Brun, T. (2025). Steganographic Entanglement Sharing. *Phys. Rev. A*, *112*, 022604

Weinstein, H., **Avritzer, B.,** Kinzfohl, C., Brun, T., and Habib, J. (2025). High Fidelity Artificial Quantum Thermal State Generation using Encoded Coherent States. *IEEE Transactions on Quantum Engineering*, vol. 6, pp. 1-7, 2025, Art no. 5100307.

Avritzer, B., & Brun, T. (2024). Quantum steganography via coherent- and Fock-state encoding in an optical medium. *Phys. Rev. A*, *109*, 032401.

Presentations and Talks

Invited Talks

“Quantum 101”. *For: Industry Network Technology Council*.

“Quantum Computing: Past, Present, Future”. *For: GMU World Quantum Day Keynote*.

“Quantum Benchmarking: US Perspective”. *For: Bitkom - Bundesverband Informationswirtschaft, Telekommunikation und neue Medien e. V.*

“Benchmarking Quantum Computing for the End User”. *For: QED-C Executive Forum*.

“An Update on Standards and Performance Metrics TAC Quantum Computing Projects” *For: QED-C Annual Meeting*.

Academic Talks

Avritzer, B., Wei, A., Clayton, C., and Cochran, R. Connecting and Correcting Multilevel Quantum Systems. *7th International Conference on Quantum Error Correction (QEC25)*.

Avritzer, B., & Brun, T. Steganographic Continuous Variable Entanglement Sharing and Quantum Communication. *APS March Meeting 2024*.

Avritzer, B., & Brun, T. Quantum Steganography using Coherent states in an Optical Channel. *Southwest Quantum Information and Technology (SQuInT) Workshop 2022*.

Service

- QED-C Standards and Performance Metrics Vice-Chair, 2026 cycle
- Reviewer, IEEE Transactions on Quantum Engineering, IEEE Transactions on Network and Service Management
- USC Jumpstart Program 2020 – provided research mentoring and career development guidance for two junior undergraduates enrolled in USC summer research programs
- Polygence Mentor – provided research mentoring and guidance along with technical help for exceptional high school students. Topics include neural network quantum decoders, theory of GKP state encodings, etc.
- Leidos Purple Buddy – provided guidance and new hire acclimation for junior employee and connected him to a research group aligned with his career goals
- Teaching Assistant: USC Physics 135b (Physics of Life Sciences), Cornell CS 1132 (Intro to Programming with Matlab) and AEP 3200 (Introductory Mathematical Physics)